

Count measure

Normal Entropy:

(in case of decision tree) $H = - \sum_{k=1}^N P_k \log_2 P_k$

average of sum of all the data sample wise

* now,

$$P_k = \frac{\sum_{i \in \text{class } k} W_i}{\sum W_i}$$

$$f_t = \underset{f}{\operatorname{argmin}} \sum_i W^{(i)} I(y^{(i)} \neq f(x^{(i)}; \theta))$$

Numerical:

x_1	x_2	y	W_i
1	2	+1	0.25
2	1	-1	0.25
4	4	+1	0.25
5	3	-1	0.25

0.144

0.144

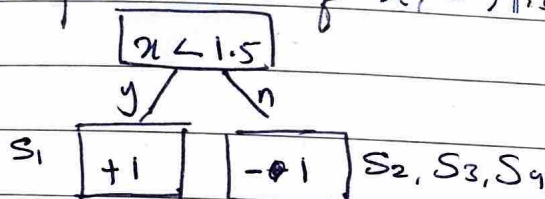
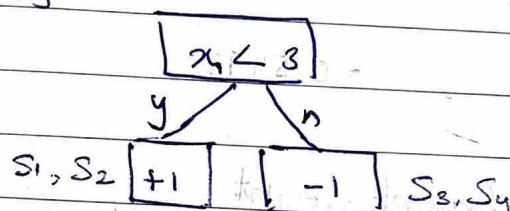
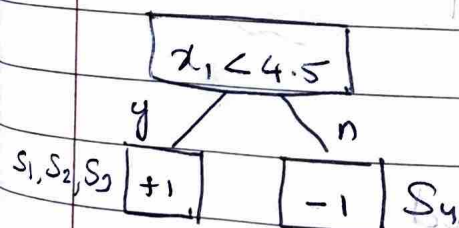
0.433

0.144

0.25 $e^{0.5493}$

estimators (break learners)

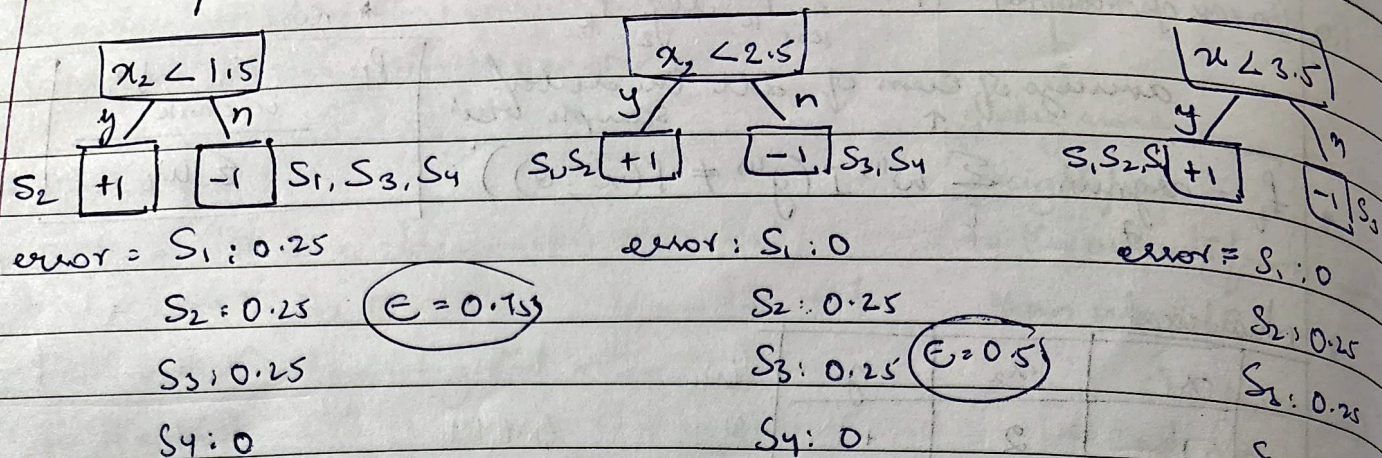
dataset (fitting can occur)

possible split values of $x_1 = \{1, 1.5, 3, 4, 5\}$ error = $S_1: 0$ $S_2: 0$ $S_3: 0.25 \rightarrow E = 0.25$ $S_4: 0$ error = $S_1: 0$ $S_2: 0.25$ $S_3: 0.25$ $S_4: 0$ $E = 0.5$ error = $S_1: 0$ $S_2: 0.25$ $S_3: 0$ $S_4: 0$ $E = 0.25$

$$\frac{1}{2} \log(1 - 0)$$

$$\frac{1}{2} \log 3 = 0.54$$

Possible split values of $x_2 = \{1.5, 2.5, 3.5\}$



2. Model weight (model chosen = $(x_1 \leq 1.5)$) f_1

$$\lambda_1 = \frac{1}{2} \log \left(\frac{0.75}{0.25} \right)$$

$$= 0.5493$$

3. Sample weight:

	old weight	New weight	Normalized
S_1	0.25	0.144	0.166
S_2	0.25	0.144	0.166
S_3	0.25	0.433	0.500
S_4	0.25	0.144	0.166

$$W^{(i)} \leftarrow W^{(i)} e^{-\lambda_1} \text{ if } f_b(x^{(i)}, \theta_b) \text{ is correct}$$

$$W^{(i)} \leftarrow W^{(i)} e^{\lambda_1} \text{ if } f_b(x^{(i)}, \theta_b) \text{ is incorrect}$$

$$S_1 = 0.25 \times e^{-0.5493} = 0.144$$

$$S_2 = 0.25 \times e^{-0.5493} = 0.144$$

$$S_3 = 0.25 \times e^{0.5493} = 0.433$$

$$S_4 = 0.25 \times e^{-0.5493} = 0.144$$

• Normalized = 0.144

$$0.144 + 0.144 + 0.433 + 0.144$$

$$= \frac{0.144}{0.865} = 0.166$$